

$$A=\pi r^2$$

$$(x+a)^n=\sum_{k=0}^n\binom{n}{k}x^ka^{n-k}$$

$$(1+x)^n=1+\frac{nx}{1!}+\frac{n(n-1)x^2}{2!}+\cdots$$

$$f(x)=a_0+\sum_{n=1}^\infty\Big(a_n\cos\frac{n\pi x}{L}+b_n\sin\frac{n\pi x}{L}\Big)$$

$$a^2+b^2=c^2$$

$$x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$$

$$e^x=1+\frac{x}{1!}+\frac{x^2}{2!}+\frac{x^3}{3!}+\cdots\,,\qquad -\infty< x<\infty$$

$$\sin\alpha\pm\sin\beta=2\sin\frac{1}{2}(\alpha\pm\beta)\cos\frac{1}{2}(\alpha\mp\beta)$$

$$\cos\alpha+\cos\beta=2\cos\frac{1}{2}(\alpha+\beta)\cos\frac{1}{2}(\alpha-\beta)$$